What Ten Industrial Surveys Identify in a Cone-Crusher Population Balance Model: Bayesian Calibration with Structural Discrepancy Gives Calibrated Bands but No Accuracy Gain over a Setting-Only Line

Felipe Santibáñez-Leal

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***Abstract*. A cone-crusher model that predicts throughput and power rests on fitted scalars, and a point fit to a few industrial surveys does not say which of them the surveys determine or how well an unseen survey is predicted. We calibrate the eight fitted scalars of a Whiten classification and breakage engine of a Metso HP500 cone crusher as a Bayesian posterior against the ten surveys of Rocha et al., with a linear structural discrepancy on the closed-side setting and the feed size in the form of Kennedy and O'Hagan, integrated out exactly and sampled with differential-evolution moves. Strict leave-one-survey-out validation refits every quantity without the held-out survey, and nine published or simple methods run on the same folds by the product. Held out, the posterior predictive mean reaches 12.00 % (95 % bootstrap interval 7.7 to 15.9) on throughput and 14.31 % (9.0 to 19.5) on power; the setting-only line reaches 12.78 % on power, and the paired difference of 1.54 points (interval −1.11 to 4.10) is no gain. The 80 % bands cover 0.8 and 0.9 of the surveys and are 43.9 % narrower on power than the residual band of the refit engine, at the width of the line's band. We prove that the engine's throughput lies in the span of the discrepancy, that the predictive mean is a ridge regression shrunk toward the engine, and that the work index and the power slope enter only as a product; a linearized posterior reproduces the sampled contraction of every parameter within 0.115. The surveys identify the capacity base and the setting dependence, not the population balance. The ablation shows that the discrepancy carries the power prediction.**

***Index Terms*. Cone crusher, population balance, Bayesian calibration, model discrepancy, identifiability, leave-one-out validation, differential-evolution MCMC, comminution.**

I. Introduction

THE electricity of a copper concentrator goes largely to crushing and grinding, and in Chile in 2024 the concentrators took 55.7 % of the electricity of copper mining [1]. The setting of a crusher decides how much work the grinding downstream must do, so a model that predicts the throughput, the power and the product size of a cone crusher from its closed-side setting and its feed is a planning tool of direct economic weight. Such models are written in the tradition of Whiten [2]: a classification function decides which particles are nipped and broken, a breakage function distributes the progeny, and a mass balance gives the product. The classification constants, the breakage law, the capacity and the power form of an engine of this kind are fitted to survey data, and the fitted values are then used as if they were known.

Industrial surveys of a crusher are few and expensive. Rocha et al. [8] report ten surveys of an HP500 secondary cone crusher on itabirite iron ore with four numbers each that are tabulated (the closed-side setting, the feed size f_{80}, the throughput and the motor power) and a product size distribution that appears only as a figure. They fitted a model of the machine on the same ten surveys. A point fit to ten rows of two measured outputs leaves two questions open. Which of the fitted scalars do the surveys determine, and which does the fit leave where it started? And how well, and with what uncertainty, is an unseen survey predicted? The published crusher models we found are calibrated by least squares or by iterative tuning and evaluated in sample (Section II); the statistical literature on computer-model calibration gives the tools to answer both questions, a posterior over the parameters and a discrepancy term that keeps the model error out of the parameters [27], [28], [30], but we found no application of it to a crusher population balance evaluated on held-out surveys.

We propose CrusherBayes, a Bayesian calibration of the engine with an explicit structural discrepancy. The engine is run as its own simulator, the eight fitted scalars receive weakly informative priors centred on the values of Rocha et al., and a discrepancy linear in the standardized setting and feed size is added to each measured output in the form of Kennedy and O'Hagan [27]. Because the discrepancy and the noise

F. Santibáñez-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanez@gmail.com; ORCID 0000-0002-0150-3246).

Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChancaDEM.

are Gaussian given the scalars, the discrepancy is integrated out exactly and the sampler explores ten dimensions. The method gives a posterior per scalar, a predictive distribution per held-out survey and a table of prior against posterior. Our contributions are:

1) the method: a Bayesian calibration with exact marginalization of a structural discrepancy for a cone-crusher population balance engine, sampled with differential-evolution moves chosen by an autocorrelation criterion;
2) a theory of what the surveys can identify: the engine's throughput lies exactly in the column space of the discrepancy design, the posterior predictive mean is a ridge regression shrunk toward the engine's best affine approximation, and the work index and the power slope enter only as a product, with a linearized posterior that reproduces the sampled contraction of all fourteen engine and discrepancy parameters;
3) a strict leave-one-survey-out comparison, with bootstrap intervals and paired differences, against nine methods run by the product on the same folds: the setting-only line, the free-form linear fit, the refit capacity base and current-based power form of the engine, the same with a bounded residual, Morrell's size-specific energy with and without a no-load term, the Bond draw, the engine at the published constants and the constant mean;
4) an ablation of the discrepancy, the engine scalars, the population-balance scalars, the prior widths, the discrepancy scale and the sampler, with three seeds of the full method.

The main result is partly negative. Held out, CrusherBayes is as accurate as the refit engine on throughput and no more accurate than a line in the closed-side setting on power, with 80 % bands whose coverage is inside 0.7 to 0.9 on both outputs. The surveys identify the capacity base, the noise and the dependence of both outputs on the setting; they leave the classification and breakage scalars of the population balance at their prior.

Section II places the method among the published approaches; Section III states the problem; Section IV gives the method; Section V proves what it can identify; Sections VI to VIII give the design, the results and the ablation, and Sections IX to XII discuss them.

II. Related Work

A. Population balance models of cone crushers

Whiten [2] modelled a crusher as a classification and breakage process solved as a mass balance; Napier-Munn et al. [3] give the breakage descriptors (t_{10} and the ore's $A \times b$) and the classification relations that such a model is calibrated with, and Andersen and Napier-Munn [4] gave a parameterization of the classification constants and a power prediction for cone crushers. Evertsson modelled the flow of material in a cone crusher [6] and its capacity and performance [7]. Bond's third theory [5] gives the specific energy of a size reduction from the feed and product P_{80}. Rocha et al. [8] fitted an Andersen–Whiten model of the HP500 to the ten surveys used here, with fitted expressions for the classification constants, the breakage parameter, the capacity and a current-based power form; we take them as the nominal values of our priors and as one of the comparators. Duarte et al. [9] calibrate a cone-crusher model with industrial data from two mines, with the entries of the breakage matrix as calibration parameters fitted by least squares. Lindqvist and Li [10] present a cone-crusher model with a coupled energy and breakage dependency, and Bengtsson [11] models energy and size distribution in cone crushers. Atta et al. [12] extend a control-oriented cone-crusher model with energy consumption and validate it on laboratory and field data, and Liu et al. [13] compute the energy of a cone crusher as the product of the compressive pressure and the displacement on the liner. Yamashita et al. [14] review the modelling and control of cone crushers. These works give the model structure and the point estimates; none of them reports a posterior of the fitted scalars or a held-out evaluation of a calibrated crusher on industrial surveys.

B. Energy and power of crushing

Morrell [15] predicts the specific energy of crushing, high-pressure grinding and tumbling mill circuits from a size-specific energy; the guideline of the Global Mining Guidelines Group [16] gives its crusher equations (B4 and B5) with the crusher work index M_{ic} of the SMC test, and defines the net power as the motor power less the no-load power. Both Bond and Morrell need a work index of the ore; the Morrell index of the Minas Rio itabirite is not published, so here it is a fitted scalar. The current-based form of Rocha et al. [8], a slope times a net power plus a no-load power, is the power model of the engine. The three forms differ in what they assume about the measured motor power: Bond's draw is a net comminution power with no idle term, Morrell's guideline subtracts a no-load power that the surveys do not report, and the current-based form fits both a slope and a no-load term to the motor power. On ten surveys each of them is therefore a fitted relation, and Section VII runs all three on the same folds.

C. Discrete-element and learned models

Discrete-element simulation with breakage is the detailed alternative to the population balance. Cleary and Sinnott [17] simulate particle flows and breakage in compression crushers, Quist and Evertsson [18] model and simulate a cone crusher with the discrete-element method, Johansson et al. [19] evaluate the performance of a cone crusher with simulations and laboratory experiments, André and Tavares [20] simulate a laboratory cone crusher with polyhedral particles, and Moncada et al. [21] predict crushing power in cone crushers with the discrete-element method. Metso report a fast-breakage discrete-element cone model developed over about a decade and regression equations fitted to it [22]. These simulations need a liner geometry and hours of computation per second of operation, and several of them are validated on laboratory machines; the HP500 liner geometry is not public. On the learned side, Koh et al. [23] use a deep network as a surrogate of the phenomenological models of a comminution circuit, Gharaei et al. [24] simulate the power of an iron-ore cone crusher

plant with a multi-method machine-learning framework, and Chen et al. [25] and Gupta and Mishra [26] solve or learn breakage population balances with physics-informed or multi-head networks. A learned model of ten surveys is a risk of overfitting; here the free-form linear fit and the setting-only line stand for this family.

D. Calibration of computer models with discrepancy

Kennedy and O'Hagan [27] calibrate the parameters of a computer model against field data by writing the data as the simulator at the true parameters plus a model discrepancy, itself a Gaussian process, plus measurement error. Higdon et al. [28] combine field data and simulations for calibration and prediction with Gaussian-process models of the unknown functions, for a simulator that cannot be run at every parameter value, and Bayarri et al. [29] give a framework for validating a computer model against reality. Brynjarsdóttir and O'Hagan [30] show that an analysis that ignores the discrepancy gives biased and over-confident parameter estimates, and that the confounding of the discrepancy with the calibration parameters is resolved only with meaningful priors. We use the low-order form of this construction: the discrepancy is linear in two standardized regressors, with a Gaussian prior on its three coefficients per output, which makes the marginal likelihood exact. Section V gives the exact form that this confounding takes for the linear discrepancy and this engine.

E. Samplers and predictive evaluation

The affine-invariant ensemble sampler of Goodman and Weare [31], implemented in `emcee` [32], moves a walker along the line through another walker (the stretch move). ter Braak [33] proposed differential-evolution Markov chain Monte Carlo, whose jumps are built from differences between chains, and ter Braak and Vrugt [34] added a snooker updater so that fewer chains suffice. Both families are gradient free, which matters here because the engine solves a linear system per evaluation in plain Python. For the evaluation of probabilistic predictions, Gneiting et al. [35] assess probabilistic forecasts by their calibration and their sharpness, and Gneiting and Raftery [36] study strictly proper scoring rules for prediction and estimation; we score the coverage, the width and the log predictive density of each held-out prediction in that frame. Yao et al. [37] average Bayesian predictive distributions by stacking, a combination we do not use because ten surveys leave too few degrees of freedom for stable weights. Our intervals on the scores use the bootstrap of Efron [38] and the score interval of Wilson [39] for a binomial proportion.

Searches in Crossref and on the open web for a Bayesian calibration with discrepancy of a crusher or mill population balance returned general methodology and no application to a cone crusher evaluated on held-out surveys; Scopus and Web of Science were not searched, so the search is not exhaustive. Among the works above, CrusherBayes takes the structure of Whiten and the point estimates of Rocha et al., the inference of Kennedy and O'Hagan with the low-order discrepancy, and the samplers of ter Braak, and adds the held-out protocol and the identifiability analysis that the crusher literature does not report.

TABLE I
THE TEN HP500 SURVEYS OF ROCHA ET AL. (TABLE A1, CC BY 4.0): CLOSED-SIDE SETTING, FEED f_{80}, THROUGHPUT AND MOTOR POWER OF EACH SURVEY.

Survey	Setting (mm)	f_{80} (mm)	Throughput (t/h)	Power (kW)
1	54	88	1639	209
2	54	84	1267	212
3	48	47	1240	262
4	48	88	1034	349
5	48	62	1124	325
6	40	79	813	355
7	38	140	1050	306
8	44	123	1138	325
9	40	81	820	350
10	55	81	1334	187

III. PROBLEM FORMULATION

A. Data

The data are the ten surveys of the Metso HP500 secondary cone crusher at an itabirite iron-ore operation of Rocha et al. [8] (Table A1 of that work, CC BY 4.0), transcribed into the product and validated by its survey contract (Table I). The survey contract parses each value, refuses one outside its bounds and requires the survey numbers 1 to 10 without a gap. The closed-side setting c spans 38 to 55 mm, the feed f_{80} 47 to 140 mm, the throughput 813 to 1639 t/h and the motor power 187 to 355 kW. Throughput and power are the two targets. The product P_{80} is not measured (the product size distributions of the source are figures only), so it is never a target; the engine's P_{80} over the ten surveys spans 21.2 to 36.6 mm and is a model output. Two correlations of the data, computed from the survey table and the engine's documents, shape every result below. The measured power falls as the setting widens (Pearson correlation -0.82), and the measured throughput rises with it (0.81); power and throughput are correlated at -0.85, while the correlation of the measured power with the engine's P_{80} is -0.14.

B. The calibration problem

Write $\mathbf{x}_i = (c_i, f_{80,i})$ for the operating point of survey i and $y_{t,i}$ for its measured output $t \in \{\mathrm{tph}, \mathrm{kW}\}$. The engine is a deterministic map $f_t(\boldsymbol{\theta}; \mathbf{x})$ from eight scalars $\boldsymbol{\theta}$ and the operating point to the output. The problem has three parts. First, a posterior $\pi(\boldsymbol{\theta} \mid \mathcal{D})$ over the scalars given the surveys $\mathcal{D}$, with a model of the error between the engine and the plant that does not force that error into the scalars. Second, for a survey h not used in the fit, a predictive distribution $p(y_{t,h} \mid \mathcal{D}_{-h})$ whose mean is accurate and whose central intervals cover the measurement at their nominal rate. Third, the identification question: for each scalar, how much the posterior narrows the prior. The assumptions are those of the engine (one breakage zone, a feed rebuilt from its f_{80}, a fixed throw and speed), that the surveys are exchangeable given the operating point, that the two outputs are independent given the scalars, and that the measurement error is Gaussian with one scale per output. The symbols are collected in Table II. The two outputs are measured; the product size, the throw and the speed are not, so an answer to the identification question is an answer for

TABLE II
NOTATION USED THROUGHOUT, WITH THE MEANING OF EACH SYMBOL AND ITS UNIT; EVERY SYMBOL IS DEFINED AGAIN AT ITS FIRST USE IN THE TEXT.

Symbol	Meaning	Unit
c, f_{80}	closed-side setting, feed 80 % passing size	mm
P_{80}, F_{80}	product and feed 80 % passing size (engine)	mm
$y_{t,i}$	measured output t of survey i	t/h or kW
$f_t(\boldsymbol{\theta};\mathbf{x})$	engine output t at scalars $\boldsymbol{\theta}$	t/h or kW
Q_{ref}	capacity base	t/h
W_i	Bond work index of the ore blend	kWh/t
ζ, P_n	slope and no-load power of the power form	–, kW
s_{K_1}, s_{K_2}, K_3	classification multipliers and exponent	–
Δt_{10}	shift of the breakage index t_{10}	points
δ_t, $\mathbf{d}_t$	discrepancy of output t and its coefficients	t/h or kW
$\mathbf{Z}$, $\mathbf{z}_i$	discrepancy design and its row	–
σ_t, τ_t	noise scale and discrepancy prior scale	t/h or kW
λ_t	shrinkage σ_t^2/τ_t^2	–
n	number of training surveys (9 in a fold)	–

these two outputs only, and a scalar the surveys cannot reach through them is reported as such rather than estimated.

IV. Proposed Method: CrusherBayes

CrusherBayes treats the crusher engine as an exact, cheap simulator and calibrates its fitted scalars jointly with an output-specific structural discrepancy. This section gives the engine, the priors, the discrepancy and its exact marginalization, the posterior predictive, the sampler, the algorithm, its cost and its implementation.

A. The engine as the simulator

For a survey with setting c and feed size f_{80} the engine evaluates a classification and breakage balance on a square-root-of-two sieve grid of 28 classes from 256 mm. The classification function is zero below K_1, one above K_2 and

$$C(d) = 1 - \left(\frac{K_2 - d}{K_2 - K_1}\right)^{K_3} \qquad (K_1 < d < K_2), \tag{1}$$

with, in millimetres and for the calibrated lane of the HP500,

$$\begin{aligned} K_1 &= s_{K_1}\,(0.23\,c + 0.30\,f_{80}),\\ K_2 &= \max\{s_{K_2}\,(12.0 + 0.55\,c + 0.40\,f_{80}),\ 1.05\,K_1\}, \end{aligned} \tag{2}$$

the coefficients being those of Rocha et al. [8] and s_{K_1}, s_{K_2} multipliers whose nominal value is one. The breakage function follows from the calibrated t_{10} relation

$$t_{10} = \min\{0.44,\ (64 + \Delta t_{10} - 0.12\,c - 0.23\,f_{80})/100\}, \tag{3}$$

through the appearance function of Austin, a two-slope cumulative distribution in the relative size that passes through t_{10} at one tenth of the parent size and is applied column by column to a strictly lower-triangular breakage matrix $\mathbf{B}$. The product mass fractions are

$$\mathbf{p} = (\mathbf{I} - \mathbf{C})(\mathbf{I} - \mathbf{B}\mathbf{C})^{-1}\mathbf{f}, \tag{4}$$

with $\mathbf{C}$ the diagonal matrix of the classification of each class and $\mathbf{f}$ the feed, rebuilt from the measured f_{80} as a Rosin–Rammler distribution of uniformity 0.9. Since $\mathbf{B}$ is strictly lower triangular, $\mathbf{I} - \mathbf{B}\mathbf{C}$ has a unit diagonal and the solve is never singular. The throughput depends on the setting and one scalar. At the fixed throw of 34 mm and speed of 360 rpm, which is the optimal speed of the machine in the capacity model, the capacity factor is the discharge opening over the reference gap of 38 mm, so

$$Q(c) = Q_{\mathrm{ref}}\,\frac{c + 17}{38}. \tag{5}$$

The Bond draw and the calibrated power are

$$\begin{aligned} P_d &= 10\,W_i\left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}}\right)Q,\\ P_c &= \zeta\,P_d + P_n, \end{aligned} \tag{6}$$

with the sizes in micrometres, W_i the Bond work index of the ore blend, and ζ and P_n the slope and the no-load power of the current-based form of Rocha et al. [8]. The eight scalars $\boldsymbol{\theta} = (Q_{\mathrm{ref}}, W_i, \zeta, P_n, s_{K_1}, s_{K_2}, K_3, \Delta t_{10})$ are the quantities calibrated. Two facts of this structure decide what the surveys can identify, and Section V makes them exact. Throughput depends on Q_{ref} alone, through (5). Power depends on all eight: on the four scalars of the population balance (s_{K_1}, s_{K_2}, K_3, Δt_{10}) only through P_{80} in the Bond term, and on W_i and ζ only through their product. The engine evaluates one operating point in about 1.3 ms, so it is run directly inside the likelihood and no emulator is needed, which removes the emulator error that Higdon et al. [28] must model.

B. Priors

Each scalar has a truncated normal prior around its nominal value, on the logarithmic scale for the positive scalars (Table III). The nominal values are those of Rocha et al. [8] (Tables 4 and 5 of that work) and, for W_i, the blend of the compact and supercompact itabirite bands (7.2 and 10.6 kWh/t, Table 1 of the same work). The widths are weakly informative choices that bracket the spans quoted there: a logarithmic standard deviation of 0.25 on Q_{ref} (about ± 28 %), 0.17 on W_i, 0.23 on ζ, 0.15 on each classification multiplier, a standard deviation of 50 kW on P_n, 0.5 on K_3 and 4 points on Δt_{10}. We found no published range of these fitted scalars for this ore, so the widths are ours, and Section VIII doubles them to measure their effect. The noise scale of each output has $\ln \sigma_t \sim N(\ln m_t, 1)$ with medians $m_{\mathrm{tph}} = 60$ t/h and $m_{\mathrm{kW}} = 15$ kW, about 5 % of the typical level of 1,200 t/h and 300 kW. Writing $\boldsymbol{\vartheta}$ for the sampled vector (the logarithm of each positive scalar, the other scalars as they are, and $\ln \sigma_t$), the log prior is

$$\ln \pi(\boldsymbol{\vartheta}) = -\sum_{j=1}^{8} \frac{(\vartheta_j - \mu_j)^2}{2 s_j^2} - \sum_{t} \frac{(\ln \sigma_t - \ln m_t)^2}{2} + \text{const}, \tag{7}$$

inside the bounds of Table III and minus infinity outside them, with μ_j and s_j the prior mean and standard deviation of the sampled scale.

TABLE III
PRIORS OF THE EIGHT ENGINE SCALARS OF THE HP500 LANE, WITH THE NOMINAL VALUE OF ROCHA ET AL., THE PRIOR ON THE SAMPLED SCALE AND ITS TRUNCATION RANGE; THE SAMPLED SCALE IS THE NATURAL LOGARITHM FOR THE POSITIVE SCALARS.

Scalar	Unit	Nominal	Prior (sampled scale)	Range
Q_{ref}	t/h	647.00	$\ln x \sim N(6.47,\, 0.25^2)$	300.00 to 1400.0
W_i	kWh/t	9.00	$\ln x \sim N(2.20,\, 0.17^2)$	5.00 to 14.0
ζ	–	1.30	$\ln x \sim N(0.26,\, 0.23^2)$	0.50 to 2.5
P_n	kW	110.00	$N(110.0,\, 50.0^2)$	0.0 to 300.0
s_{K_1}	–	1.00	$\ln x \sim N(0.00,\, 0.15^2)$	0.50 to 2.0
s_{K_2}	–	1.00	$\ln x \sim N(0.00,\, 0.15^2)$	0.50 to 2.0
K_3	–	2.30	$N(2.3,\, 0.5^2)$	1.0 to 4.0
Δt_{10}	points of t_{10}	0.00	$N(0.0,\, 4.0^2)$	−15.0 to 15.0

C. Discrepancy and exact marginal likelihood

For survey i and output t,

$$y_{t,i} = f_t(\boldsymbol{\theta}; \mathbf{x}_i) + \delta_t(\mathbf{z}_i) + \varepsilon_{t,i}, \quad \varepsilon_{t,i} \sim N(0, \sigma_t^2), \tag{8}$$

$$\delta_t(\mathbf{z}) = d_{t,0} + d_{t,1} z_c + d_{t,2} z_{f_{80}}, \quad d_{t,j} \sim N(0, \tau_t^2), \tag{9}$$

where $\mathbf{z} = (z_c, z_{f_{80}})$ is the pair of setting and feed size standardized with the mean and the population standard deviation of the training surveys, and $\tau_{\mathrm{tph}} = 140$ t/h and $\tau_{\mathrm{kW}} = 36$ kW (about 12 % of the typical level). The discrepancy is the model-form error of Kennedy and O'Hagan [27] reduced to a plane in the two regressors that the surveys vary: a constant that absorbs a level error of the engine and two slopes that absorb a wrong dependence on the setting or the feed size. Given $\boldsymbol{\theta}$ and σ_t the model is linear and Gaussian in $\mathbf{d}_t$, so the discrepancy is integrated out exactly. With $\mathbf{Z}$ the $n \times 3$ matrix of rows $(1, z_{c,i}, z_{f_{80},i})$ of the training surveys and $\mathbf{r}_t = \mathbf{y}_t - \mathbf{f}_t(\boldsymbol{\theta})$ the residual of the engine,

$$\mathbf{y}_t \sim N(\mathbf{f}_t(\boldsymbol{\theta}),\ \boldsymbol{\Sigma}_t), \qquad \boldsymbol{\Sigma}_t = \sigma_t^2 \mathbf{I} + \tau_t^2 \mathbf{Z}\mathbf{Z}^\top. \tag{10}$$

The two outputs are treated as independent given $\boldsymbol{\theta}$, so the log likelihood is the sum of the two Gaussian log densities of (10), computed through the Cholesky factor of $\boldsymbol{\Sigma}_t$; the unnormalized log posterior is

$$\ln \pi(\boldsymbol{\vartheta} \mid \mathcal{D}) = \ln \pi(\boldsymbol{\vartheta}) + \text{const} - \sum_t \left[\tfrac{1}{2} \mathbf{r}_t^\top \boldsymbol{\Sigma}_t^{-1} \mathbf{r}_t + \tfrac{1}{2} \ln \det \boldsymbol{\Sigma}_t \right]. \tag{11}$$

The sampler explores the eight scalars and the two noise scales (ten dimensions); the six discrepancy coefficients are never sampled. For each retained sample they follow from the exact Gaussian conditional

$$\mathbf{d}_t \mid \boldsymbol{\theta}, \sigma_t, \mathcal{D} \sim N(\mathbf{m}_t, \mathbf{S}_t),$$
$$\mathbf{m}_t = \tau_t^2 \mathbf{Z}^\top \boldsymbol{\Sigma}_t^{-1} \mathbf{r}_t, \quad \mathbf{S}_t = \tau_t^2 \mathbf{I} - \tau_t^4 \mathbf{Z}^\top \boldsymbol{\Sigma}_t^{-1} \mathbf{Z}, \tag{12}$$

which is how the identifiability table reports the discrepancy coefficients. The low-order form is a choice for nine training surveys: a Gaussian-process discrepancy as in Kennedy and O'Hagan [27] would add a length scale and an amplitude per output to be learned from nine points in two dimensions, whereas the plane has three coefficients per output, a closed-form marginal and a conditional that the identifiability table can report coefficient by coefficient. The price is that a curved model error in the setting is not represented, and it appears as noise in σ_t.

D. Posterior predictive of a held-out survey

For a survey h outside the training set, with $\mathbf{z}_h$ standardized by the training mean and deviation and $\mathbf{x}_h^\top = (1, z_{c,h}, z_{f_{80},h})$, the predictive at one retained sample is Gaussian,

$$y_{t,h} \mid \boldsymbol{\theta}, \sigma_t, \mathcal{D}_{-h} \sim N\big(f_t(\boldsymbol{\theta}; \mathbf{x}_h) + \mathbf{x}_h^\top \mathbf{m}_t,\ \sigma_t^2 + \mathbf{x}_h^\top \mathbf{S}_t \mathbf{x}_h\big), \tag{13}$$

and the posterior predictive is the mixture of these Gaussians over the retained samples. Its mean is the average of the component means. The central 80 % interval is the 10th to the 90th percentile of 20 draws per retained sample from (13), and the predictive density of the measurement is the average of the component densities, which is the quantity the log score of Section VI uses. The predictive carries three sources of uncertainty: the noise σ_t, the conditional uncertainty of the discrepancy $\mathbf{S}_t$, and the spread of the scalars across the samples. It is a generative model of a survey, which is what distinguishes its band from a residual band built from the spread of held-out errors. The discrepancy at the held-out survey uses the standardization of the training surveys, so a held-out setting outside the training range extrapolates the plane linearly.

E. Sampler and convergence criterion

The sampler is `emcee` 3.1.6 [32] with the differential-evolution move of ter Braak [33] and the snooker move of ter Braak and Vrugt [34], chosen at random in proportions 0.8 and 0.2. The walkers are updated in groups (two for the differential-evolution move, four for the snooker move), each group proposing from the walkers of the others. In the installed implementation, the differential-evolution proposal of a walker $\boldsymbol{\vartheta}$ is

$$\boldsymbol{\vartheta}^* = \boldsymbol{\vartheta} + \gamma\,(\boldsymbol{\vartheta}_a - \boldsymbol{\vartheta}_b), \qquad \gamma = \gamma_0(1 + 10^{-5}\xi),\ \gamma_0 = \frac{2.38}{\sqrt{2D}}, \tag{14}$$

with $\boldsymbol{\vartheta}_a \neq \boldsymbol{\vartheta}_b$ two walkers of the other group, $\xi \sim N(0,1)$ and $D = 10$ the dimension, and the proposal is accepted with probability $\min\{1, \pi(\boldsymbol{\vartheta}^* \mid \mathcal{D})/\pi(\boldsymbol{\vartheta} \mid \mathcal{D})\}$. The snooker move projects the difference of two other walkers on the line through the walker and a third one, with a stretch of 1.7. Both moves use only the log posterior (11), no gradient. The run has 20 walkers (twice the dimension), 7000 steps, the first 1000 discarded and every 10th kept, which retains 12000 samples per posterior, from a fixed seed (42). The walkers start in a ball around the prior centre of 5 % of each prior standard deviation. A posterior counts as converged when

$$T - T_{\mathrm{burn}} \ \geq\ 50\,\hat{\tau}_{\max}, \tag{15}$$

with T the steps, T_{burn} the discarded steps and $\hat{\tau}_{\max}$ the largest over the ten dimensions of the integrated autocorrelation time that `emcee` estimates. A posterior that fails (15) is recorded as not converged and never kept silently. The affine-invariant stretch move of Goodman and Weare [31], the default of `emcee`, was the first choice and fails this criterion on the same posterior (Section VIII).

F. Algorithm, cost and implementation

Algorithm 1 gives one posterior and the predictive of its held-out survey. The cost is dominated by the engine. One log posterior evaluates the engine on the n training surveys (two outputs per evaluation) and factors two $n \times n$ matrices, so a posterior costs about WTn engine evaluations, here $20 \times 7000 \times 9 = 1.26 \times 10^6$, and the predictive step adds $(n+1)$ evaluations for each of the 12000 retained samples. The linear algebra is $O(n^3)$ per evaluation with $n = 9$ and is negligible beside the engine's 28-class balance. The eleven posteriors of the protocol (ten folds and the fit to all ten surveys) are independent, and the walkers of all of them are evaluated in parallel by a pool of processes; the document does not depend on the number of processes, which a test of the product checks. The implementation is the module `data-pipeline/pipeline/hp500/crusherbayes.py` of the product, on the canonical Python engine `data-pipeline/pipeline/model/crusher/`, which takes the eight scalars through one optional argument whose defaults are the shipped values; the TypeScript port of the engine that runs in the browser is untouched. The parameters of the method are the prior widths, the discrepancy scales τ_t, the noise prior, the sampler size and the seed, all listed above and in the committed document, and none of them was tuned on the held-out scores. The number of walkers is twice the sampled dimension, and the number of steps was sized to the convergence criterion: on the same posterior the differential-evolution moves give an autocorrelation time of about 50 steps, so 6,000 post-burn steps hold more than the 50 autocorrelation times the criterion asks. The predictive step evaluates the engine once per retained sample at the training surveys and the held-out survey, and the discrepancy conditional (12) needs one more $n \times n$ inverse per sample and output. The bake writes one document for the whole protocol, the folds with their predictive summaries, the convergence record, the scores of the comparators, the contraction table and the terms of the rule for dropping the method, and the product's contract tests validate it.

Algorithm 1 CrusherBayes: one posterior and the predictive of the held-out survey h

Require: training surveys $\mathcal{D}_{-h}$, held-out operating point $\mathbf{x}_h$, priors, τ_t, seed
standardize (c, f_{80}) on $\mathcal{D}_{-h}$; build $\mathbf{Z}$ and $\mathbf{x}_h$
draw W walkers in a ball of 5 % of the prior widths around the prior centre
for $k = 1, \dots, T$ **do**
  for each group of the walkers **do**
    propose by (14) (probability 0.8) or the snooker move (0.2)
    evaluate the engine on $\mathcal{D}_{-h}$; $\ell \leftarrow$ (11) via Cholesky of $\mathbf{\Sigma}_t$
    accept or reject
  end for
end for
estimate $\hat{\tau}_{\max}$; record converged if (15) holds
keep every 10th sample after T_{burn}
for each retained sample **do**
  compute $\mathbf{m}_t, \mathbf{S}_t$ by (12); the engine at $\mathbf{x}_h$
  store mean and variance of (13); draw 20 predictive values
end for
return predictive mean, 10th and 90th percentiles, mixture density at $y_{t,h}$

V. Theory

This section states why the method behaves as it does on these surveys: what the data can and cannot tell about each scalar, and what form the predictive mean takes. Propositions 2 to 4 are exact for the model of Section IV; Proposition 1 is exact for the parts of the engine that are linear in a scalar and a first-order statement for the others, and Conjecture 1 is supported by the evidence given with it.

A. The information a discrepancy leaves to the scalars

Let $\mathbf{Z} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$ be the thin singular value decomposition of the design, with $\mathbf{U}$ an $n \times 3$ matrix of orthonormal columns and $\mathbf{S} = \mathrm{diag}(s_1, s_2, s_3)$; over the ten surveys the singular values span 2.44 to 3.75. Since $\mathbf{Z}\mathbf{Z}^\top = \mathbf{U}\mathbf{S}^2\mathbf{U}^\top$,

$$\mathbf{\Sigma}_t^{-1} = \frac{1}{\sigma_t^2}\left(\mathbf{I} - \mathbf{U}\mathbf{U}^\top\right) + \mathbf{U}\left(\sigma_t^2\mathbf{I} + \tau_t^2\mathbf{S}^2\right)^{-1}\mathbf{U}^\top, \quad (16)$$

which one checks by multiplying by $\mathbf{\Sigma}_t = \sigma_t^2(\mathbf{I} - \mathbf{U}\mathbf{U}^\top) + \mathbf{U}(\sigma_t^2\mathbf{I} + \tau_t^2\mathbf{S}^2)\mathbf{U}^\top$: the two terms act on orthogonal subspaces.

Proposition 1: Let $\mathbf{j} = \partial\mathbf{f}_t/\partial\vartheta$ be the sensitivity of output t on the training surveys to a sampled scalar ϑ, and $\mathbf{P} = \mathbf{U}\mathbf{U}^\top$ the projection on the column space of $\mathbf{Z}$. The Fisher information of the marginal likelihood (10) about ϑ is

$$\mathcal{I}_t(\vartheta) = \frac{\|(\mathbf{I} - \mathbf{P})\mathbf{j}\|^2}{\sigma_t^2} + \sum_{k=1}^{3} \frac{(\mathbf{u}_k^\top \mathbf{j})^2}{\sigma_t^2 + \tau_t^2 s_k^2}. \quad (17)$$

Proof: For a Gaussian with mean $\mathbf{f}_t(\vartheta)$ and a covariance that does not depend on ϑ, the Fisher information is $\mathbf{j}^\top\mathbf{\Sigma}_t^{-1}\mathbf{j}$. Inserting (16) and writing $\mathbf{j} = (\mathbf{I} - \mathbf{P})\mathbf{j} + \mathbf{U}\mathbf{U}^\top\mathbf{j}$ gives the two terms. The result is exact when $\mathbf{f}_t$ is linear in ϑ and holds to first order otherwise. ∎

The proposition says that a discrepancy linear in (c, f_{80}) discounts exactly the part of a sensitivity that is itself an affine function of the setting and the feed size, by the factor $\sigma_t^2/(\sigma_t^2 + \tau_t^2 s_k^2)$, and leaves the orthogonal part untouched. A scalar is identified by the surveys only through the part of its effect that a plane in the two regressors cannot imitate, or through the finite width of the discrepancy prior. With the noise scales at their posterior means on all ten surveys and the values of τ_t, the discount factor for a singular value of 3 is about one tenth on both outputs.

B. Throughput and the capacity base

Proposition 2: On the HP500 lane the vector of engine throughputs of the training surveys lies in the column space of $\mathbf{Z}$, for every value of the scalars: $\mathbf{f}_{\mathrm{tph}}(\boldsymbol{\theta}) = Q_{\mathrm{ref}}\,\mathbf{Z}\mathbf{a}$ with $\mathbf{a} = ((\bar{c}+17)/38,\ s_c/38,\ 0)^\top$, $\bar{c}$ and s_c the training mean and

deviation of the setting. Consequently the throughput surveys carry no information about the other seven scalars, and their information about $\vartheta = \ln Q_{\mathrm{ref}}$ is

$$\mathcal{I}_{\mathrm{tph}}(\ln Q_{\mathrm{ref}}) = Q_{\mathrm{ref}}^2\, \mathbf{a}^\top \mathbf{Z}^\top \mathbf{Z} \left(\sigma^2 \mathbf{I} + \tau^2 \mathbf{Z}^\top \mathbf{Z}\right)^{-1} \mathbf{a} \leq \frac{Q_{\mathrm{ref}}^2 \|\mathbf{a}\|^2}{\tau^2}, \tag{18}$$

with σ, τ the throughput scales, so that it vanishes as $\tau \to \infty$.

Proof: By (5), $Q_i = Q_{\mathrm{ref}}(c_i + 17)/38$, and $c_i = \bar{c} + s_c z_{c,i}$, so $Q_i = Q_{\mathrm{ref}}[(\bar{c} + 17)/38 + (s_c/38) z_{c,i}]$, which is row i of $Q_{\mathrm{ref}} \mathbf{Z}\mathbf{a}$. None of the other scalars enters (5). Then $\mathbf{j} = Q_{\mathrm{ref}} \mathbf{Z}\mathbf{a}$ has no component orthogonal to the column space and, by (17) with $\mathbf{Z} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, $\mathcal{I} = Q_{\mathrm{ref}}^2 \mathbf{a}^\top \mathbf{V}\mathbf{S}^2 (\sigma^2 \mathbf{I} + \tau^2 \mathbf{S}^2)^{-1} \mathbf{V}^\top \mathbf{a}$, which is (18). Each factor $s_k^2/(\sigma^2 + \tau^2 s_k^2)$ is below $1/\tau^2$, which gives the bound. ∎

Computed with the product's engine on the ten surveys, the relative norm of the component of the capacity factors outside the column space of $\mathbf{Z}$ is 2.9×10^{-16}, which is the proposition at machine precision. The capacity base is therefore not identified by the shape of the throughput data, which a plane in the setting reproduces whatever Q_{ref} is; it is identified only because the discrepancy prior penalizes a large plane. The posterior contraction of Q_{ref} must therefore move toward one when τ_{tph} grows and toward smaller values when it shrinks.

C. The predictive mean is a shrunk regression

Proposition 3: Write $\lambda_t = \sigma_t^2/\tau_t^2$, $\mathbf{G} = \mathbf{Z}^\top \mathbf{Z}$, and let $\mathbf{b}_t$ be the least-squares coefficients of the engine output on $\mathbf{Z}$, $\mathbf{f}_t = \mathbf{Z}\mathbf{b}_t + \mathbf{e}_t$ with $\mathbf{Z}^\top \mathbf{e}_t = \mathbf{0}$, and $e_{t,h} = f_t(\boldsymbol{\theta}; \mathbf{x}_h) - \mathbf{x}_h^\top \mathbf{b}_t$ the engine's departure from that plane at the held-out survey. Given the scalars, the predictive mean of (13) is

$$\hat{y}_{t,h} = \mathbf{x}_h^\top \left(\mathbf{G} + \lambda_t \mathbf{I}\right)^{-1} \left(\mathbf{Z}^\top \mathbf{y}_t + \lambda_t \mathbf{b}_t\right) + e_{t,h}. \tag{19}$$

Proof: The push-through identity $\mathbf{Z}^\top (\sigma^2 \mathbf{I} + \tau^2 \mathbf{Z}\mathbf{Z}^\top)^{-1} = (\sigma^2 \mathbf{I} + \tau^2 \mathbf{G})^{-1} \mathbf{Z}^\top$ turns (12) into $\mathbf{m}_t = (\mathbf{G} + \lambda_t \mathbf{I})^{-1} \mathbf{Z}^\top (\mathbf{y}_t - \mathbf{f}_t)$. With $\mathbf{Z}^\top \mathbf{f}_t = \mathbf{G}\mathbf{b}_t$, the predictive mean $f_{t,h} + \mathbf{x}_h^\top \mathbf{m}_t = \mathbf{x}_h^\top \mathbf{b}_t + e_{t,h} + \mathbf{x}_h^\top (\mathbf{G} + \lambda_t \mathbf{I})^{-1} (\mathbf{Z}^\top \mathbf{y}_t - \mathbf{G}\mathbf{b}_t)$, and $\mathbf{b}_t - (\mathbf{G} + \lambda_t \mathbf{I})^{-1} \mathbf{G}\mathbf{b}_t = \lambda_t (\mathbf{G} + \lambda_t \mathbf{I})^{-1} \mathbf{b}_t$ gives (19). ∎

Corollary 1: As $\tau_t \to \infty$ ($\lambda_t \to 0$) the predictive mean tends to the ordinary least-squares fit of the survey on the standardized setting and feed size plus the engine's non-affine remainder $e_{t,h}$; for throughput $e_{t,h} = 0$ by Proposition 2, so the limit is the free-form linear fit. As $\lambda_t \to \infty$ it tends to the engine. Between them, the engine enters through a shrinkage target whose weight on each standardized coefficient is $\lambda_t/(g_{kk} + \lambda_t)$ for a diagonal $\mathbf{G}$.

The corollary bounds what CrusherBayes can achieve on these data: given the scalars, its predictive mean is a blend of a linear fit in the two regressors and the engine's best plane, plus the engine's curvature. The values in use are $\lambda_{\mathrm{tph}} = 1.08$ and $\lambda_{\mathrm{kW}} = 1.16$ with the noise at its posterior mean on all ten surveys, so the weight of the engine's plane is about 0.10 on both outputs. On power the engine's best plane over the ten surveys has a slope of 8.5 kW per standard deviation of the setting, against -56.0 kW for the measured power, so the shrinkage pulls the slope toward the wrong sign; the method cannot beat a linear fit unless the curvature $e_{t,h}$ carries signal.

D. The work index and the power slope

Proposition 4: The engine power depends on W_i and ζ only through ζW_i, so the likelihood is constant along the curves $\ln \zeta + \ln W_i = \text{const}$, and

$$\frac{\partial \mathbf{f}_{\mathrm{kW}}}{\partial \ln W_i} = \frac{\partial \mathbf{f}_{\mathrm{kW}}}{\partial \ln \zeta} = \zeta\, \mathbf{P}_d. \tag{20}$$

The data inform $\ln \zeta + \ln W_i$ and nothing else about the pair; the marginal of each is the prior along the ridge reweighted by the information on the sum.

Proof: In (6), P_d is linear in W_i, and $P_c = \zeta W_i \cdot 10(P_{80}^{-1/2} - F_{80}^{-1/2})Q + P_n$; P_{80} does not depend on W_i, which on the calibrated lane enters only the Bond term. Differentiating in the logarithms gives the same vector ζP_d. ∎

The finite-difference Jacobians of the engine in $\ln W_i$ and $\ln \zeta$ agree to a relative difference of 9.1×10^{-14}. The no-load power P_n enters power as a constant, $\partial \mathbf{f}_{\mathrm{kW}}/\partial P_n = \mathbf{1}$, which lies in the column space of $\mathbf{Z}$; by Proposition 1 it is discounted like the capacity base and is confounded with the constant of the power discrepancy. The linearized posterior (21) below gives a correlation of -0.27 between W_i and ζ: weaker than the exact ridge suggests, because the two priors have different widths and the information on their sum is small beside the prior precision, so the ridge reweights the prior only slightly.

E. The population balance

Conjecture 1: On these surveys, with throughput and power as the only measured outputs, the four population-balance scalars are not identified: their posterior equals their prior up to the Monte Carlo error of the sampler, and fixing them at their nominal value changes neither the held-out accuracy nor the bands beyond that error.

The evidence is threefold. First, by Proposition 2, throughput carries nothing about them. Second, they reach power only through P_{80} in the Bond term, and the Bond draw of the engine varies only from 196 to 237 kW over the ten surveys, against a measured span of 187 to 355 kW. Third, the share of the power sensitivity of each scalar that lies in the column space of $\mathbf{Z}$, and is therefore discounted by Proposition 1, is at least 0.906, the lowest being that of Δt_{10} (Table IV). A second-order statement for all fourteen parameters follows from a Laplace approximation of the linearized model: with $\mathbf{A}_t$ the Jacobian of the mean of output t in the eight sampled scalars and the six discrepancy coefficients, and $\boldsymbol{\Lambda}$ the diagonal prior precision,

$$\mathrm{Cov} \approx \left(\boldsymbol{\Lambda} + \sum_t \sigma_t^{-2} \mathbf{A}_t^\top \mathbf{A}_t\right)^{-1}, \tag{21}$$

whose diagonal, divided by the prior variance, predicts the ratio of posterior to prior standard deviation. Evaluated at the posterior mean of the fit to all ten surveys, it predicts 0.457 for Q_{ref}, 0.920 for W_i, 0.848 for ζ, 0.852 for P_n and 0.92 to 0.99 for the four population-balance scalars, and it agrees with the ratio the sampler measured within 0.115 for every one of the fourteen parameters (Table IV). The largest differences are on ζ and W_i, where the truncated,

TABLE IV
THEORY AGAINST THE SAMPLER ON THE FIT TO ALL TEN SURVEYS: THE SHARE OF EACH SCALAR'S POWER SENSITIVITY INSIDE THE COLUMN SPACE OF THE DISCREPANCY DESIGN, THE RATIO OF POSTERIOR TO PRIOR STANDARD DEVIATION PREDICTED BY THE LINEARIZED POSTERIOR (21), AND THE RATIO THE SAMPLER MEASURED.

Parameter	Share of the power sensitivity in the design span	Linearized ratio	Sampled ratio	Difference
Q_{ref}	0.999	0.457	0.454	0.003
W_i	0.999	0.920	0.854	0.066
ζ	0.999	0.848	0.733	0.115
P_n	1.000	0.852	0.795	0.057
s_{K_1}	0.976	0.918	0.965	−0.047
s_{K_2}	0.990	0.980	1.026	−0.046
K_3	0.991	0.994	0.992	0.002
Δt_{10}	0.906	0.994	0.990	0.004
$d_{\mathrm{tph},0}$	–	0.859	0.878	−0.019
$d_{\mathrm{tph},1}$ (setting)	–	0.343	0.350	−0.007
$d_{\mathrm{tph},2}$ (f_{80})	–	0.335	0.340	−0.005
$d_{\mathrm{kW},0}$	–	0.926	0.910	0.016
$d_{\mathrm{kW},1}$ (setting)	–	0.401	0.381	0.020
$d_{\mathrm{kW},2}$ (f_{80})	–	0.359	0.362	−0.003

logarithmic prior and the curved ridge of Proposition 4 make the Gaussian approximation coarsest. The conjecture is tested by the ablation that fixes the four scalars (Section VIII).

F. Conditions

The propositions hold under conditions that the HP500 lane meets and another machine or data set may not. The exactness of Proposition 2 needs a capacity that is affine in the setting, which holds because the throw and the speed are fixed and the speed is at the optimum of the capacity model; with a varying speed the capacity hump of the engine is not affine and the throughput would carry information outside the span. Proposition 3 needs the discrepancy to be linear in the regressors with independent Gaussian coefficients, so a Gaussian-process discrepancy as in Kennedy and O'Hagan [27] would replace the ridge by a kernel smoother with the same structure. Conjecture 1 depends on the outputs measured: a measured product size distribution would give the population-balance scalars a sensitivity that a plane in the setting cannot absorb, and would change it. Identification of the capacity base and the no-load power rests on the finiteness of τ_t, which is the formal content of the statement of Brynjarsdóttir and O'Hagan [30] that the confounding is resolved only by meaningful priors.

VI. EXPERIMENTAL DESIGN

A. Hypotheses stated before the experiments

Four hypotheses were written before the posteriors were computed, two of them as the rule for dropping the method written into the product's design document, and two of them as the risks of the method in the research that preceded it.

H1 Accuracy. The posterior predictive mean beats the setting-only line on power by at least 3.0 points of mean absolute percentage error, or beats the refit capacity base on throughput by the same margin.

H2 Calibration and sharpness. On both outputs the coverage of the 80 % interval lies within 0.7 to 0.9 and the mean width of the interval is no larger than that of the residual band of the refit engine.

H3 Identifiability. With ten surveys and two outputs the posterior of most scalars is dominated by the prior.

Algorithm 2 Strict leave-one-survey-out protocol, shared by every method

Require: ten surveys, a method $\mathcal{M}$ that fits on surveys and predicts one

for $h = 1, \ldots, 10$ **do**
  $\mathcal{D}_{-h} \leftarrow$ the nine other surveys
  fit $\mathcal{M}$ on $\mathcal{D}_{-h}$ only (scalars, noise, standardization)
  predict survey h: mean $\hat{y}_h$ and, if $\mathcal{M}$ has one, an 80 % interval
end for
for a residual-band method, $\sigma_h \leftarrow$ sample deviation of the held-out residuals of the nine surveys $\neq h$
score MAPE, RMSE, log density, coverage and width over the ten surveys
bootstrap the ten surveys 20000 times (seed 42) for intervals and paired differences
return the scores with their intervals

H4 Source of the gain. Most of what the method gains over the refit engine comes from the discrepancy, which the setting-only line already approximates.

The method is dropped as a predictor if H1 fails and H2 fails, both together; whatever the outcome, the identifiability table is reported. With $n = 10$ a difference below about 3.0 points of MAPE is within the noise. The propositions of Section V add consequences that the ablation tests: the capacity-base contraction moves with τ_t, the power accuracy moves toward the linear fit as τ_t grows, and fixing the population-balance scalars changes nothing beyond the Monte Carlo error.

B. Data, folds and protocol

The dataset is the ten HP500 surveys of Section III, used as published, with no survey removed. The protocol is strict leave-one-survey-out (Algorithm 2): for each survey a posterior is computed from the other nine alone, and the engine scalars, the noise scales, the discrepancy and the standardization of the regressors are all refit inside the fold, so the held-out survey enters only the prediction. A test of the product plants an absurd value in one survey and checks that every other fold moves and its own does not. One more posterior from all ten surveys gives the identifiability table, making 11 posteriors. Every comparator is refit inside the same folds on the same nine surveys, so all methods are compared on identical training and test sets. The rule for dropping the method is evaluated by code from the scores alone, and a test of the product checks it in every combination of its terms.

C. Published methods compared

All comparators are computed by the product's code on the same folds, from its committed held-out document; none of them is a reported number of another work. For both outputs: the setting-only line, an ordinary least-squares line of the output on the closed-side setting; the free-form linear fit, an ordinary least-squares fit on the setting and the feed size; the constant mean of the nine training surveys; and the refit engine of Rocha et al., the engine at the nominal scalars with the

capacity base refit by least squares for throughput and the slope and intercept of the current-based form $P_c = \zeta P_d + P_n$ refit for power, with the same refit plus a ridge residual on the standardized regressors bounded to $\pm 20\,\%$ of the engine value. For power only: Morrell's size-specific energy [15], [16] with the crusher work index M_{ic} refit per fold through the origin, and the same with a refit no-load term, both on the measured f_{80}, the measured throughput and the engine's P_{80}. Two references fit nothing on the surveys: the bare Bond draw at the blend work index, and the engine at the published constants of Rocha et al., whose constants those authors fitted on the same ten surveys, so that this row is an in-sample figure of their fit and not a held-out one. The residual band of a comparator is its prediction plus or minus $1.2816\,\sigma_h$, with σ_h the sample standard deviation of the held-out residuals of the other nine surveys, so that a survey never enters the width of its own band.

D. Metrics

For output t, held-out predictions $\hat{y}_{t,i}$ and measurements $y_{t,i}$ over the $N = 10$ surveys,

$$\mathrm{MAPE}_t = \frac{100}{N}\sum_{i=1}^{N}\frac{|\hat{y}_{t,i} - y_{t,i}|}{y_{t,i}},$$

$$\mathrm{RMSE}_t = \Big(\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_{t,i} - y_{t,i})^2\Big)^{1/2}. \tag{22}$$

The log predictive density is the mean over surveys of the log density of the measurement under the held-out predictive,

$$\mathrm{LPD}_t = \frac{1}{N}\sum_{i=1}^{N}\ln\Big[\frac{1}{M}\sum_{m=1}^{M}\phi\big(y_{t,i};\mu_i^{(m)},v_i^{(m)}\big)\Big], \tag{23}$$

with ϕ the Gaussian density and $(\mu_i^{(m)}, v_i^{(m)})$ the mean and variance of (13) at retained sample m; for a comparator it is the Gaussian density with its residual-band deviation. It is a strictly proper score [36], in log units per t/h or per kW, higher being better. The coverage and the mean width of the 80 % interval $[\ell_{t,i}, u_{t,i}]$ are

$$\kappa_t = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\{\ell_{t,i} \le y_{t,i} \le u_{t,i}\},$$

$$w_t = \frac{1}{N}\sum_{i=1}^{N}(u_{t,i} - \ell_{t,i}), \tag{24}$$

read together as the calibration and the sharpness of the interval [35]. The identifiability of a parameter is the ratio

$$\rho_j = \mathrm{sd}_{\mathrm{post}}(\theta_j)/\mathrm{sd}_{\mathrm{prior}}(\theta_j) \tag{25}$$

on the natural scale, read as identified up to 0.5, partly identified up to 0.9 and at its prior above 0.9. MAPE and RMSE measure the point prediction, the log density measures the whole predictive distribution, and coverage with width measure the interval; a method can be good on one and poor on another, which is why the hypotheses H1 and H2 are stated on different metrics. The prior standard deviation of a positive scalar is computed on the natural scale by sampling its truncated log-normal prior.

E. Statistics

The ten held-out surveys are the unit of resampling. The 95 % interval of a MAPE or an RMSE is the percentile interval of 20000 bootstrap resamples of the ten surveys [38], with a fixed seed (42); a paired difference of MAPE between CrusherBayes and a comparator is bootstrapped with the same resamples, so that the correlation of the two errors across surveys is kept, and reported with the fraction of resamples in which CrusherBayes has the lower MAPE and the number of surveys on which it has the lower error. The interval of a coverage k/N is the Wilson score interval [39],

$$\frac{\hat{p} + \frac{z^2}{2N} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \qquad z = 1.96. \tag{26}$$

The posterior uncertainty of each parameter is its posterior standard deviation. The Monte Carlo error of the sampler is measured by repeating the whole protocol with the seeds 43 and 44. With ten surveys these intervals are wide, and they are reported so that a difference is read against them rather than as a ranking of point values. No significance test is used to accept or reject a hypothesis.

F. Ablations planned

Each variant changes one component and reruns the whole protocol (eleven posteriors) with the product's code, the same folds, sampler size and seed. The discrepancy: its scale reduced to 10^{-6} of τ_t, so the likelihood is the engine plus noise. The engine scalars: all eight held at their nominal value (prior width 10^{-4} of its value), so that only the discrepancy and the noise are learned. The shared population-balance scalars: s_{K_1}, s_{K_2}, K_3 and Δt_{10} held at their nominal value while capacity, work index and power form are sampled. The priors: every engine prior width doubled. The discrepancy scale: τ_t halved and doubled. The sampler: the stretch move in place of the differential-evolution moves. The seeds 43 and 44 of the full method measure the Monte Carlo error against which every variant is read. Each variant writes a document of the same contract as the method, so that the same scores, the same contraction table and the same rule for dropping the method are computed for it by the same code.

VII. Results

A. Held-out accuracy against the published methods

Table V gives every method on the same folds with its bootstrap interval, and Fig. 1 draws the intervals. On throughput the posterior predictive mean has a MAPE of 12.00 % (95 % interval 7.7 to 15.9), against 12.09 % for the refit capacity base, 12.18 % for the free-form fit, 13.45 % with the bounded residual, 13.67 % for the setting-only line and 18.29 % for the constant mean. The engine at the published constants reaches 11.55 %, but it is the fit of Rocha et al. to the same ten surveys, so it is not a held-out figure. The intervals of the first five methods overlap almost entirely. On power the posterior predictive mean has a MAPE of 14.31 % (9.0 to 19.5), against 12.78 % (7.4 to 18.4) for the setting-only line, 14.03 % for the free-form fit and 15.22 % for Morrell with a no-load term. The

engine-based power models are far behind: 25.94 % for the refit current-based form, 25.37 % with the bounded residual, 26.88 % for the Bond draw, 38.08 % for Morrell through the origin and 47.58 % for the engine at the published constants, against 24.48 % for the constant mean.

The paired comparison (Table VI) settles H1. Against the setting-only line on power the difference of MAPE is 1.54 points (95 % interval −1.11 to 4.10), CrusherBayes has the lower error on 5 of the ten surveys, and the gain the hypothesis asked for, 3.0 points, lies outside the interval. Against the refit capacity base on throughput the difference is −0.09 points (−6.30 to 6.06), a tie. H1 does not hold. The comparison does show where the method gains: against every engine-based power model the difference is large and its interval excludes zero, −11.63 points against the refit current-based form (−18.25 to −4.47) and −12.56 points against the Bond draw (−21.98 to −3.58), and against the constant mean on power −10.16 points (−19.32 to −0.72). The gain is over the engine, not over the simple lines.

B. Calibration and sharpness

The 80 % interval of CrusherBayes covers 0.8 of the surveys on throughput (Wilson interval 0.49 to 0.94) and 0.9 on power (0.60 to 0.98), both inside 0.7 to 0.9. Its mean width is 439.12 t/h on throughput, 7.3 % narrower than the residual band of the refit engine (473.76 t/h) at the same coverage, and 117.74 kW on power, 43.9 % narrower than that band (209.79 kW) at the same coverage of 0.9. H2 holds. Against the simple models the picture is different: on power the setting-only line has a mean width of 119.60 kW at a coverage of 0.8, and the free-form fit 129.82 kW at 0.8, so the band of CrusherBayes has about the width of the line's band with a coverage higher by one survey. On throughput all four banded methods cover 0.8 and CrusherBayes has the narrowest band. The Wilson intervals of the coverages, 0.60 to 0.98 and 0.49 to 0.94, overlap almost entirely: with ten surveys the coverages do not distinguish a calibrated band from a slightly miscalibrated one. The log predictive density (Table VII) agrees: −5.35 on power for CrusherBayes against −5.34 for the line and −5.85 for the refit engine. Figure 2 shows the bands survey by survey, and Fig. 3 draws them against the setting.

Figure 3 makes the structure of Section V visible. The engine at the published constants puts the power at 365 to 419 kW, rising with the setting, while the measured power falls from 355 to 187 kW as the setting widens; the held-out mean of CrusherBayes follows the measurements, because the discrepancy has taken over the setting dependence. On throughput the engine, the line and CrusherBayes all follow the rise of the throughput with the setting, and the scatter around it, which neither the setting nor the feed size explains, sets the width of every band.

C. The dropping rule

Table VIII applies the rule stated before the run. Condition (a) holds: the gain in power MAPE over the setting-only line is −1.53 points and the gain in throughput MAPE over the refit capacity base is 0.09 points, both below 3.0. Condition (b) fails: the coverages 0.8 and 0.9 are inside the interval 0.7 to 0.9, and the bands are narrower than the residual band on both outputs. Both conditions are needed to drop the method, so the rule does not drop it. The rule therefore retains a method that calibrates its band and does not improve the point prediction. Every variant of the ablation gives the same outcome: condition (a) holds and condition (b) fails in all of them, including the variant without the discrepancy, whose coverage of 0.7 is at the lower edge of the interval and whose band remains narrower than the residual band.

D. What the surveys identify

Table IX compares the prior with the posterior of every parameter, fitted to all ten surveys, and Figs. 4 and 5 draw it. Of the 16 parameters, 7 are identified, 4 are partly identified and 5 stay at their prior. The identified ones are the capacity base Q_{ref} (ratio 0.454), the two noise scales (0.301 and 0.332) and the four discrepancy slopes (0.340 to 0.381). Partly identified are ζ (0.733), P_n (0.795), W_i (0.854) and the constant of the throughput discrepancy (0.878). Four of the eight engine scalars stay at their prior: s_{K_1} (0.965), s_{K_2} (1.026), K_3 (0.992) and Δt_{10} (0.990), and so does the constant of the power discrepancy (0.910). The posterior of Q_{ref} is 646.35 ± 75.74 t/h against a nominal 647 t/h: the surveys narrow the capacity base and leave its centre where it was. H3 holds, in the specific form that the theory predicts: the scalars that only power can inform, through a sensitivity that lies almost wholly in the span of the discrepancy, stay at their prior, and the linearized posterior of Table IV reproduces every ratio.

E. The discrepancy and the shrinkage

The discrepancy slope on the setting is large for both outputs. For power it is −57.07 ± 13.70 kW per standard deviation of the setting (about −9.6 kW/mm, with 5.9 mm the standard deviation of the setting over the ten surveys), and for throughput 100.84 ± 49.00 t/h per standard deviation (about 17.0 t/h per mm). The measured power falls as the setting widens (correlation −0.82), whereas the engine power at the published constants rises with it (correlation 0.79, from 365 to 419 kW); the negative slope of the power discrepancy compensates the setting dependence of the engine. Figure 6 evaluates the closed form (19) with the scalars at their nominal value over the shrinkage λ. On throughput the held-out MAPE is 12.18 % at $\lambda = 0$, which is the free-form fit as Corollary 1 states, and 11.53 % for a large λ, which is the engine at the published constants; between them it dips to 10.78 % near $\lambda = 17.8$, and at the λ of the method it is 11.91 %, against 12.00 % for the sampled method. A stronger shrinkage would have fit these folds better, but choosing λ on the held-out scores would tune the method on its test set, and the dip of about one point is inside the bootstrap interval of every throughput method. On power it rises from 13.12 % at $\lambda = 0$ to 47.17 % for a large λ, with 15.39 % at the λ of the method against 14.31 % sampled: shrinking toward the engine costs accuracy on power, as its wrong-signed slope predicts.

TABLE V
HELD-OUT SCORES OF CRUSHERBAYES AND OF EVERY COMPARED METHOD ON THE TEN HP500 SURVEYS (STRICT LEAVE-ONE-SURVEY-OUT, THE SAME FOLDS FOR ALL), WITH 95 % PERCENTILE BOOTSTRAP INTERVALS OF THE MAPE OVER THE SURVEYS, AND THE COVERAGE OF THE 80 % INTERVAL WITH ITS WILSON INTERVAL AND THE MEAN WIDTH FOR THE METHODS THAT GIVE A BAND; THE ENGINE AT THE PUBLISHED CONSTANTS IS IN SAMPLE FOR THE FIT OF ROCHA ET AL.

Target	Method	Fitted on the training surveys	MAPE (%)	95% interval	RMSE	Coverage [95% interval]	Width
Power (kW)							
	CrusherBayes	eight scalars, noise, discrepancy	14.31	[9.0, 19.5]	44.7	0.9 [0.60, 0.98]	117.7
	Setting-only line	intercept, slope	12.78	[7.4, 18.4]	44.6	0.8 [0.49, 0.94]	119.6
	Free-form linear fit	intercept, two slopes	14.03	[8.4, 19.7]	48.7	0.8 [0.49, 0.94]	129.8
	Morrell with no-load term	M_{ic}, no-load power	15.22	[6.4, 24.9]	46.5	–	–
	Refit current-based form	ζ, P_n	25.94	[17.3, 34.9]	78.5	0.9 [0.60, 0.98]	209.8
	Refit form with bounded residual	ζ, P_n, ridge residual	25.37	[16.0, 37.9]	92.1	–	–
	Morrell through the origin	M_{ic}	38.08	[24.4, 54.1]	111.7	–	–
	Constant mean	the mean	24.48	[15.2, 35.2]	68.5	–	–
	Bond draw, blend work index	none	26.88	[20.0, 33.2]	93.2	–	–
	Engine at the published constants	none; published fit	47.58	[24.6, 73.5]	132.8	–	–
Throughput (t/h)							
	CrusherBayes	eight scalars, noise, discrepancy	12.00	[7.7, 15.9]	168.6	0.8 [0.49, 0.94]	439.1
	Refit capacity base	Q_{ref}	12.09	[6.0, 18.7]	176.7	0.8 [0.49, 0.94]	473.8
	Free-form linear fit	intercept, two slopes	12.18	[7.6, 16.3]	174.4	0.8 [0.49, 0.94]	468.5
	Capacity base with bounded residual	Q_{ref}, ridge residual	13.45	[8.6, 17.7]	189.5	–	–
	Setting-only line	intercept, slope	13.67	[9.7, 18.0]	177.8	0.8 [0.49, 0.94]	478.8
	Constant mean	the mean	18.29	[9.2, 28.5]	258.5	–	–
	Engine at the published constants	none; published fit	11.55	[7.4, 16.2]	171.4	–	–

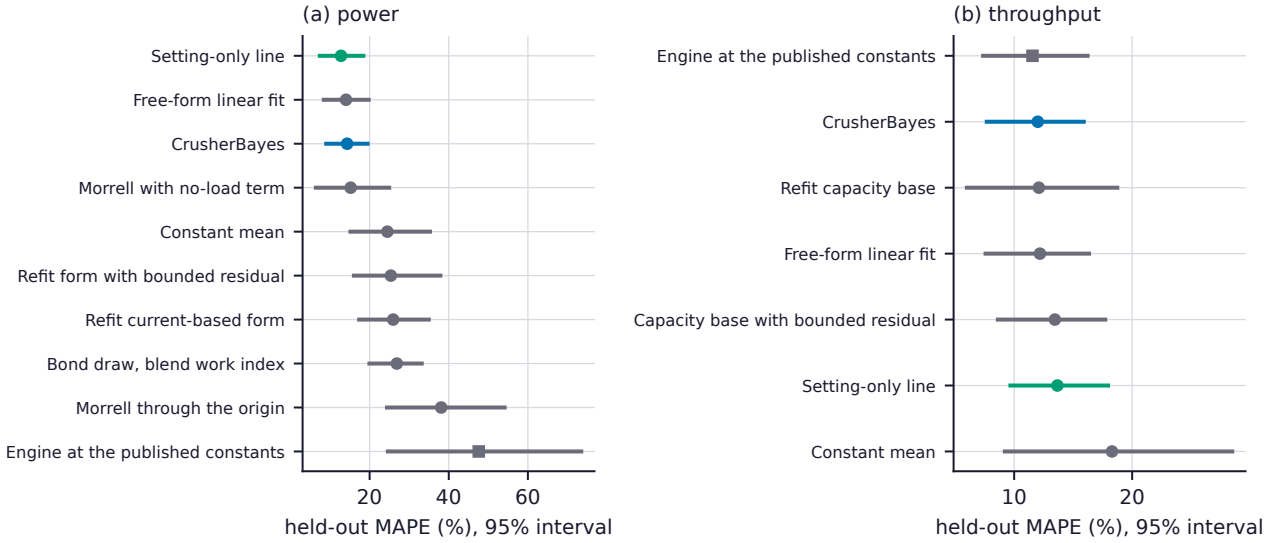


Fig. 1. Held-out mean absolute percentage error of every method on the ten HP500 surveys with its 95 % bootstrap interval, for (a) power and (b) throughput; CrusherBayes in blue, the setting-only line in green, the engine at the published constants (in sample for its fit) as a square.

F. Convergence

All 11 posteriors meet the criterion (15) (Table X and Fig. 7): the largest autocorrelation time is between 48.0 and 84.8 steps and the 6000 post-burn steps are 70.7 to 125.0 times it, with an acceptance fraction between 0.171 and 0.204. The fold without survey 4 has the slowest chain. Across the three seeds the throughput MAPE varies by 0.01 points and the power MAPE by 0.02 points, and the power band width by 0.1 kW, so the Monte Carlo error of the sampler is two orders of magnitude below the differences between methods. The retained samples of a fold, 12000, are therefore far more than the summaries need: the posterior means and standard deviations of the identifiability table and the percentiles of the predictive interval are averages over at least 70 effective draws per walker, and the ratios of posterior to prior standard deviation of the three seeds differ by at most 0.02.

VIII. ABLATION

Table XI and Fig. 8 give the ablation; every row is a full rerun of the eleven posteriors with the product's code. The seed rows bound the Monte Carlo error: 11.99 and 11.99 % on throughput and 14.29 and 14.31 % on power, against 12.00 and 14.31 % for seed 42.

The discrepancy carries the power prediction. Without it, the power MAPE rises to 27.15 %, back to the level of the refit engine (25.94 %), the band widens to 190.21 kW and

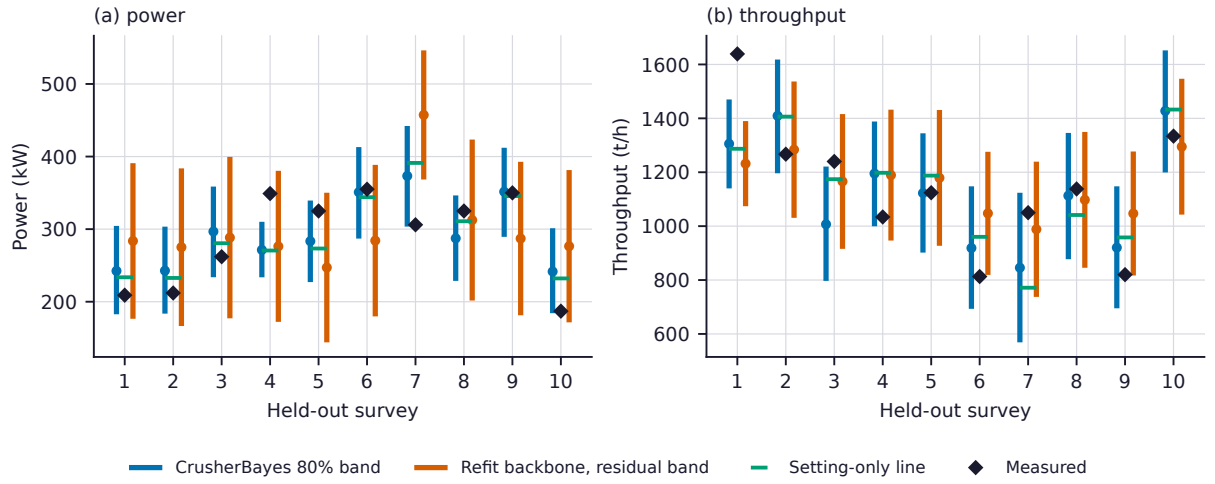


Fig. 2. Held-out 80 % predictive interval against the measured value, per survey, for (a) power and (b) throughput: the posterior predictive of CrusherBayes in blue, the residual band of the refit engine in orange, the setting-only line as a green dash and the measurement as a black diamond; every interval is computed without the survey it is drawn for.

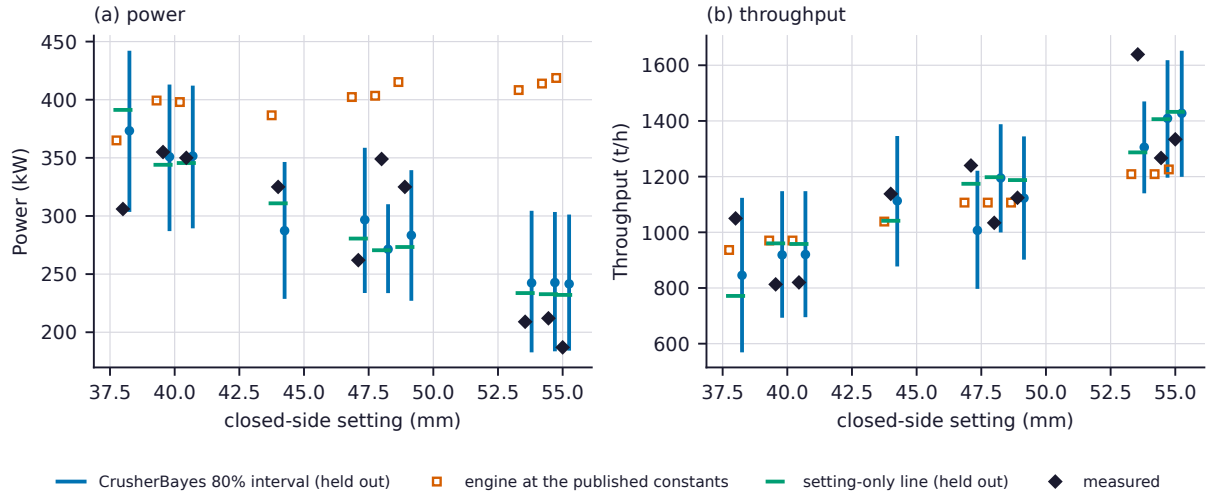


Fig. 3. The two outputs of the ten surveys against the closed-side setting: the measurement, the held-out interval and mean of CrusherBayes, the held-out setting-only line and the engine at the published constants, for (a) power and (b) throughput; surveys at one setting are spread by 0.9 mm.

TABLE VI
PAIRED DIFFERENCES OF HELD-OUT MAPE, CRUSHERBAYES MINUS EACH COMPARATOR (NEGATIVE FAVOURS CRUSHERBAYES), WITH THE 95 % PAIRED-BOOTSTRAP INTERVAL, THE FRACTION OF RESAMPLES FAVOURING CRUSHERBAYES AND THE SURVEYS ON WHICH IT HAS THE LOWER ERROR.

Target	Comparator	Δ MAPE (points)	95% interval	$P(\Delta < 0)$	Surveys won
Power					
	Setting-only line	1.54	[−1.11, 4.10]	0.130	5 of 10
	Free-form linear fit	0.28	[−3.36, 3.77]	0.432	6 of 10
	Morrell with no-load term	−0.90	[−8.71, 6.89]	0.584	5 of 10
	Refit current-based form	−11.63	[−18.25, −4.47]	0.999	7 of 10
	Refit form with bounded residual	−11.06	[−21.30, −3.63]	1.000	8 of 10
	Morrell through the origin	−23.77	[−39.44, −9.70]	1.000	9 of 10
	Constant mean	−10.16	[−19.32, −0.72]	0.983	6 of 10
	Bond draw, blend work index	−12.56	[−21.98, −3.58]	0.998	7 of 10
	Engine at the published constants	−33.27	[−56.55, −12.16]	1.000	8 of 10
Throughput					
	Refit capacity base	−0.09	[−6.30, 6.06]	0.503	5 of 10
	Free-form linear fit	−0.18	[−1.36, 1.01]	0.623	6 of 10
	Capacity base with bounded residual	−1.45	[−2.06, −0.76]	1.000	9 of 10
	Setting-only line	−1.67	[−4.61, 2.23]	0.825	8 of 10
	Constant mean	−6.29	[−15.85, 2.19]	0.913	5 of 10
	Engine at the published constants	0.45	[−3.44, 4.49]	0.416	6 of 10

TABLE VII
HELD-OUT SCORES OF THE METHODS THAT GIVE AN 80 % BAND, OVER THE TEN SURVEYS: MAPE, RMSE, MEAN LOG PREDICTIVE DENSITY, COVERAGE AND MEAN WIDTH OF THE INTERVAL, FOR EACH OUTPUT.

Target	Method	MAPE (%)	RMSE	Mean log density	Coverage	Width
Throughput (t/h)						
	CrusherBayes	12.00	168.57	−6.678	0.8	439.1
	Setting-only line	13.67	177.75	−6.709	0.8	478.8
	Refit backbone with residual band	12.09	176.67	−6.896	0.8	473.8
	Free-form linear fit	12.18	174.42	−6.647	0.8	468.5
Power (kW)						
	CrusherBayes	14.31	44.68	−5.346	0.9	117.7
	Setting-only line	12.78	44.56	−5.341	0.8	119.6
	Refit backbone with residual band	25.94	78.52	−5.847	0.9	209.8
	Free-form linear fit	14.03	48.68	−5.399	0.8	129.8

the coverage falls to 0.7 on both outputs; the log density of power drops from −5.35 to −5.82. Throughput hardly changes (11.98 %), because the engine throughput is already a line in the setting (Proposition 2); the capacity base contracts more without the discrepancy (ratio 0.195 against 0.454), as

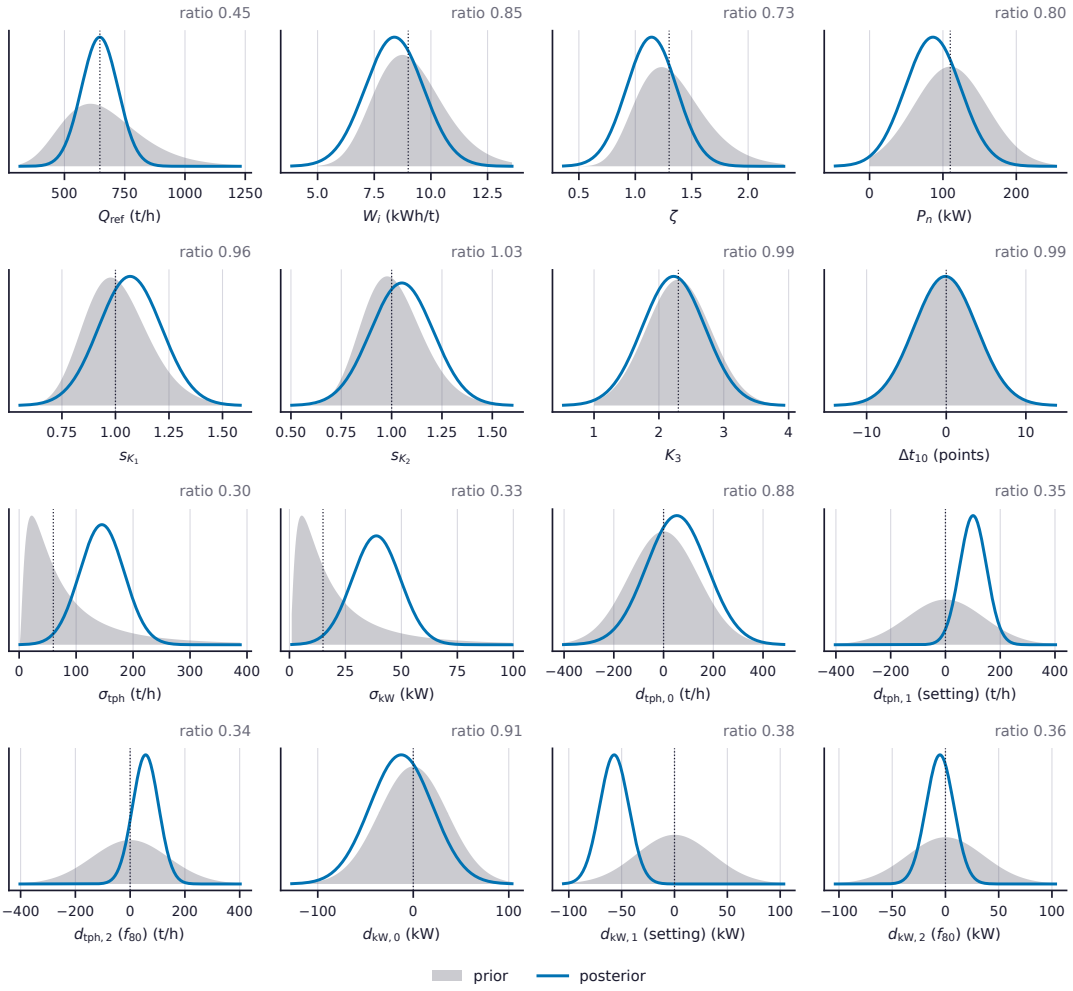


Fig. 4. Prior (grey area) against posterior (blue line) of each of the sixteen parameters, fitted to all ten surveys; the posterior is drawn as a normal density with its mean and standard deviation, the dotted line is the nominal value and the ratio is the posterior over the prior standard deviation.

TABLE VIII
THE TERMS OF THE RULE FOR DROPPING THE METHOD AS A PREDICTOR, EVALUATED ON THE HELD-OUT SCORES OF THE TEN SURVEYS; THE REFERENCE IS THE SETTING-ONLY LINE FOR POWER AND THE REFIT CAPACITY BASE FOR THROUGHPUT.

Term	Throughput	Power
MAPE of CrusherBayes (%)	12.00	14.31
MAPE of the reference (%)	12.09	12.78
Gain over the reference (points)	0.09	−1.53
Gain of at least 3 points	no	no
Coverage	0.8	0.9
Coverage outside 0.7 to 0.9	no	no
Mean width (t/h, kW)	439.1	117.7
Mean width of the residual band	473.8	209.8
Band wider than the residual band	no	no

Proposition 2 predicts when $\tau_{\mathrm{tph}} \to 0$. H4 holds.

The engine scalars contribute little. With all eight held at their nominal value the throughput MAPE is 11.95 % and the power MAPE 15.99 %, 15.99 against 14.31 %, with a power band of 124.50 kW at a coverage of 0.7; calibrating the scalars improves power by less than two points and does not change throughput. The discrepancy slopes are the same with and

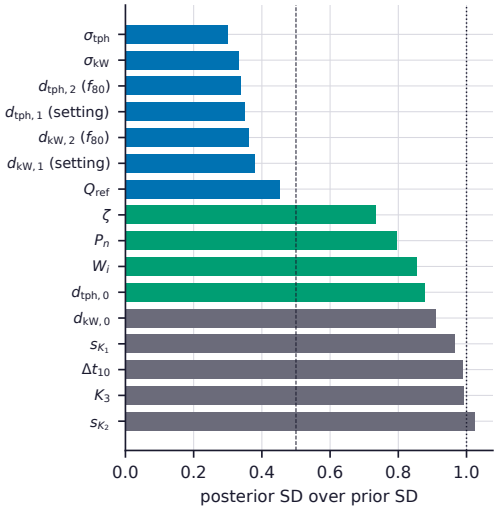


Fig. 5. Posterior over prior standard deviation of each parameter on the fit to all ten surveys: identified (blue, up to 0.5), partly identified (green, up to 0.9) and at the prior (grey), with dashed and dotted lines at 0.5 and 1.

TABLE IX

PRIOR AGAINST POSTERIOR OF THE 16 PARAMETERS OF CRUSHERBAYES, FITTED TO ALL TEN HP500 SURVEYS; THE RATIO IS THE POSTERIOR STANDARD DEVIATION OVER THE PRIOR STANDARD DEVIATION ON THE NATURAL SCALE, AND THE READING FOLLOWS THE THRESHOLDS 0.5 AND 0.9.

Parameter	Unit	Prior mean	Prior SD	Posterior mean	Posterior SD	Ratio	Reading
Q_{ref}	t/h	647.00	167.01	646.35	75.74	0.454	identified
W_i	kWh/t	9.00	1.52	8.39	1.29	0.854	partly identified
ζ	–	1.30	0.30	1.14	0.22	0.733	partly identified
P_n	kW	110.00	47.85	86.35	38.04	0.795	partly identified
s_{K_1}	–	1.00	0.15	1.07	0.15	0.965	at its prior
s_{K_2}	–	1.00	0.15	1.05	0.16	1.026	at its prior
K_3	–	2.30	0.49	2.23	0.49	0.992	at its prior
Δt_{10}	points	0.00	3.99	−0.11	3.95	0.990	at its prior
σ_{tph}	t/h	60.00	129.67	145.23	39.04	0.301	identified
σ_{kW}	kW	15.00	32.42	38.81	10.75	0.332	identified
$d_{\mathrm{tph},0}$	t/h	0.00	140.00	53.36	122.88	0.878	partly identified
$d_{\mathrm{tph},1}$ (setting)	t/h	0.00	140.00	100.84	49.00	0.350	identified
$d_{\mathrm{tph},2}$ (f_{80})	t/h	0.00	140.00	56.57	47.63	0.340	identified
$d_{\mathrm{kW},0}$	kW	0.00	36.00	−12.57	32.77	0.910	at its prior
$d_{\mathrm{kW},1}$ (setting)	kW	0.00	36.00	−57.07	13.70	0.381	identified
$d_{\mathrm{kW},2}$ (f_{80})	kW	0.00	36.00	−5.16	13.03	0.362	identified

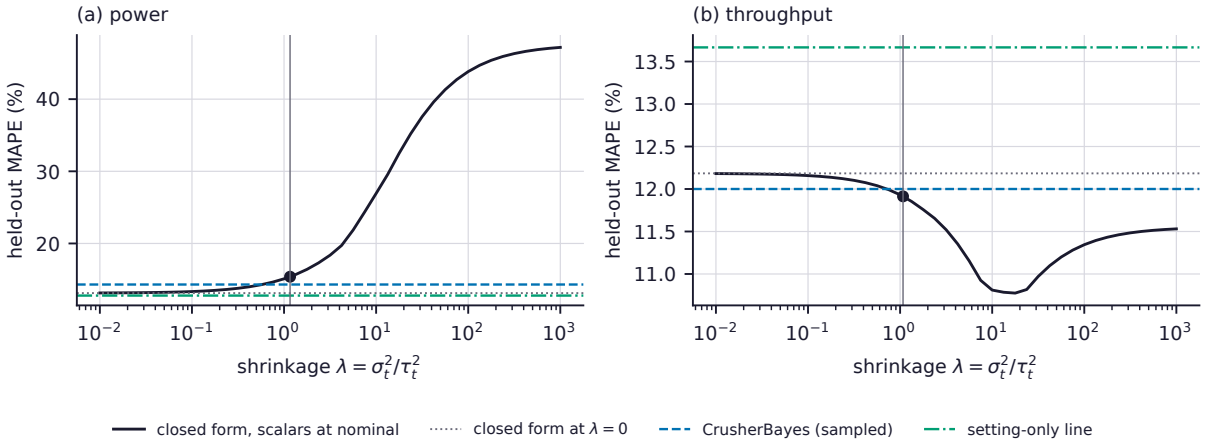


Fig. 6. Held-out MAPE of the closed-form predictive mean (19) with the engine scalars at their nominal value, against the shrinkage λ, for (a) power and (b) throughput; the dot marks the λ of the method, the dashed line the sampled CrusherBayes and the dash-dotted line the setting-only line.

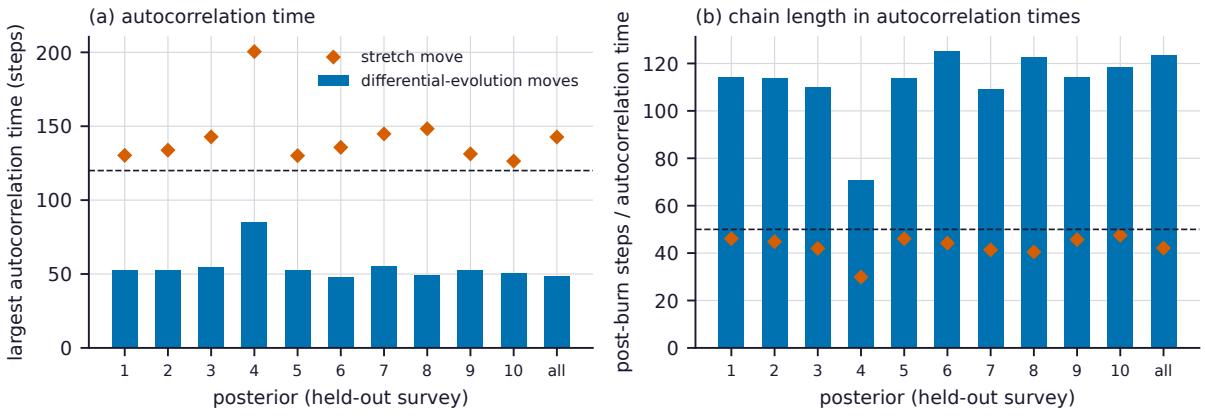


Fig. 7. Convergence of the eleven posteriors (ten folds and all ten surveys): (a) the largest integrated autocorrelation time and (b) the post-burn chain length in autocorrelation times, for the differential-evolution moves (bars) and the stretch move (diamonds); the dashed lines mark the criterion, an autocorrelation time of at most 120 steps in (a) and a chain of at least 50 autocorrelation times in (b).

without the scalars (−55.8 against −57.1 kW on power). The population-balance scalars contribute little. Held at their

TABLE X
CONVERGENCE OF THE 11 POSTERIORS OF THE METHOD: THE LARGEST INTEGRATED AUTOCORRELATION TIME $\tau_{\max}$ OVER THE TEN DIMENSIONS, THE POST-BURN CHAIN LENGTH IN UNITS OF $\tau_{\max}$ (THE CRITERION IS 50), THE MEAN ACCEPTANCE FRACTION AND THE RETAINED SAMPLES.

Posterior	$\tau_{\max}$ (steps)	Post-burn steps / $\tau_{\max}$	Acceptance	Retained samples
without survey 1	52.6	114.1	0.204	12000
without survey 2	52.7	113.8	0.201	12000
without survey 3	54.7	109.7	0.203	12000
without survey 4	84.8	70.7	0.171	12000
without survey 5	52.7	113.9	0.203	12000
without survey 6	48.0	125.0	0.204	12000
without survey 7	55.1	108.9	0.200	12000
without survey 8	48.9	122.7	0.204	12000
without survey 9	52.6	114.1	0.203	12000
without survey 10	50.7	118.3	0.202	12000
all ten surveys	48.6	123.5	0.204	12000

nominal value, they leave the throughput MAPE at 11.99 %, the power MAPE at 14.08 % and the power band at 117.70 kW, all close to the full method, and the contraction of the other scalars hardly moves (Q_{ref} 0.444, ζ 0.687). The power MAPE is 0.23 points lower than with the four scalars sampled, a difference larger than the seed spread of 0.02 points, so sampling them has a real but small effect through the engine's curvature, far inside the survey-level uncertainty of several points. The identification part of Conjecture 1 stands; its part on the scores holds at the level of the survey noise and not at the level of the Monte Carlo error.

The priors matter for identification and not for accuracy. Doubling every engine prior width gives 12.08 % and 14.43 %, with a power band of 119.24 kW; the ratios are taken against the wider prior, so Q_{ref} falls to 0.314 while W_i (0.889) and ζ (0.776) remain partly identified, and one fold of eleven (without survey 4) fails the convergence criterion, with an autocorrelation time of 141.7 steps: the wider prior opens the ridge of Proposition 4.

The discrepancy scale moves the method along the shrinkage of Proposition 3. Halving τ_t shrinks harder toward the engine: power degrades to 17.85 % with a band of 134.98 kW, and the capacity base contracts to 0.294. Doubling τ_t shrinks less: power improves to 13.85 %, closer to the linear limit of Corollary 1, and the capacity base loosens to 0.638, as (18) states. Throughput moves by less than a quarter of a point either way. None of the variants changes the outcome of the dropping rule.

The sampler changes the diagnostics and not the scores. The stretch move gives 12.00 % and 14.29 %, equal to the differential-evolution moves within the seed spread, but none of the eleven posteriors meets the criterion: its autocorrelation time is 126.4 to 200.5 steps, so the post-burn chain is below 50 autocorrelation times in every posterior (Fig. 7). The choice of the move is a choice of reliability of the posterior summaries, which are the identifiability result.

IX. DISCUSSION

A. Why the point accuracy does not improve

Proposition 3 answers it. Given the scalars, the predictive mean is a ridge regression of the survey on the standardized setting and feed size, shrunk with a weight of about one tenth toward the engine's best plane, plus the engine's non-affine remainder. On throughput the engine's plane is the engine itself (Proposition 2), and the method is a blend of the free-form fit (12.18 %) and the refit capacity line (12.09 %); it lands between and ties both, as the paired differences show. On power the engine's plane has the wrong sign of the setting slope, so shrinking toward it costs accuracy, and the closed form traces the cost as λ grows (Fig. 6); the remainder $e_{t,h}$ is small because almost all of the engine's power variation is affine in the two regressors. A line in the setting alone already captures the strongest dependence of the data, and with ten surveys a second regressor costs more in variance than it gains. The negative result on power is therefore structural, not a failure of the sampler, and it would persist with any amount of computation on these data.

B. Why the population balance is not identified

The two measured outputs carry the product size only through the Bond term of power, and the Bond draw of the engine spans only 196 to 237 kW across surveys whose power spans 187 to 355 kW. More than nine tenths of the power sensitivity of each population-balance scalar is an affine function of the setting and the feed size, which the discrepancy absorbs at a discount of about one tenth (Proposition 1); what is left is below the noise. The four scalars therefore stay at their prior, the engine's product size is not validated by these surveys, and the ablation that fixes them changes the scores by far less than the survey-level noise. Measured product size distributions per survey would constrain them directly; the distributions of the source are figures only, so this was not tested. Proposition 1 also says how much such data would help: a measured size has a sensitivity to the classification constants that is not an affine function of the setting and the feed size, so it would be counted at the full weight $1/\sigma^2$ rather than at the discount of the plane.

C. Confounding of calibration and discrepancy

Brynjarsdóttir and O'Hagan [30] show that the discrepancy is confounded with the calibration parameters. Here the confounding has an exact form. The capacity base is identified only through the finite width of the throughput discrepancy prior, and its contraction moves with τ_{tph} from 0.294 to 0.638 when the scale is halved or doubled. The no-load power is a constant, confounded with the constant of the power discrepancy, and both are only partly identified or at the prior. The work index and the power slope enter as a product, and each of them is only partly identified. The posterior of an individual scalar should therefore not be read as an estimate of a physical value of the ore or the machine: W_i at 8.39 $\pm$ 1.29 kWh/t is the prior reweighted along a ridge, not a measurement of the work index of the itabirite. The same holds for ζ and P_n, whose posterior means of 1.14 and 86.35 kW are the values that, together with the discrepancy, reproduce ten power measurements.

D. What the band is worth, and the negative results

The band of CrusherBayes comes from a generative model of a survey (engine, discrepancy and noise), so it carries the

TABLE XI
ABLATION OF CRUSHERBAYES ON THE TEN HP500 SURVEYS: EVERY ROW RERUNS THE ELEVEN POSTERIORS OF THE STRICT LEAVE-ONE-SURVEY-OUT PROTOCOL WITH ONE COMPONENT CHANGED, AND GIVES THE HELD-OUT MAPE, COVERAGE AND MEAN WIDTH OF EACH OUTPUT, THE RANGE OF THE LARGEST AUTOCORRELATION TIME AND THE CONVERGED POSTERIORS.

Variant	Throughput			Power			$\tau_{\max}$	Converged
	MAPE	Cov.	Width	MAPE	Cov.	Width	(steps)	
The method (seed 42)	12.00	0.8	439	14.31	0.9	117.7	48–85	11 of 11
The method, seed 43	11.99	0.8	438	14.29	0.9	117.8	50–80	11 of 11
The method, seed 44	11.99	0.8	439	14.31	0.9	117.7	47–73	11 of 11
Without the discrepancy	11.98	0.7	451	27.15	0.7	190.2	54–76	11 of 11
Engine scalars fixed at nominal	11.95	0.8	425	15.99	0.7	124.5	34–39	11 of 11
Population-balance scalars fixed	11.99	0.8	439	14.08	0.8	117.7	46–54	11 of 11
Engine prior widths doubled	12.08	0.8	440	14.43	0.9	119.2	60–142	10 of 11
Discrepancy scale τ_t halved	11.89	0.9	435	17.85	0.8	135.0	49–108	11 of 11
Discrepancy scale τ_t doubled	12.12	0.8	446	13.85	0.8	117.3	47–89	11 of 11
Stretch move instead of differential evolution	12.00	0.8	440	14.29	0.9	117.8	126–200	0 of 11

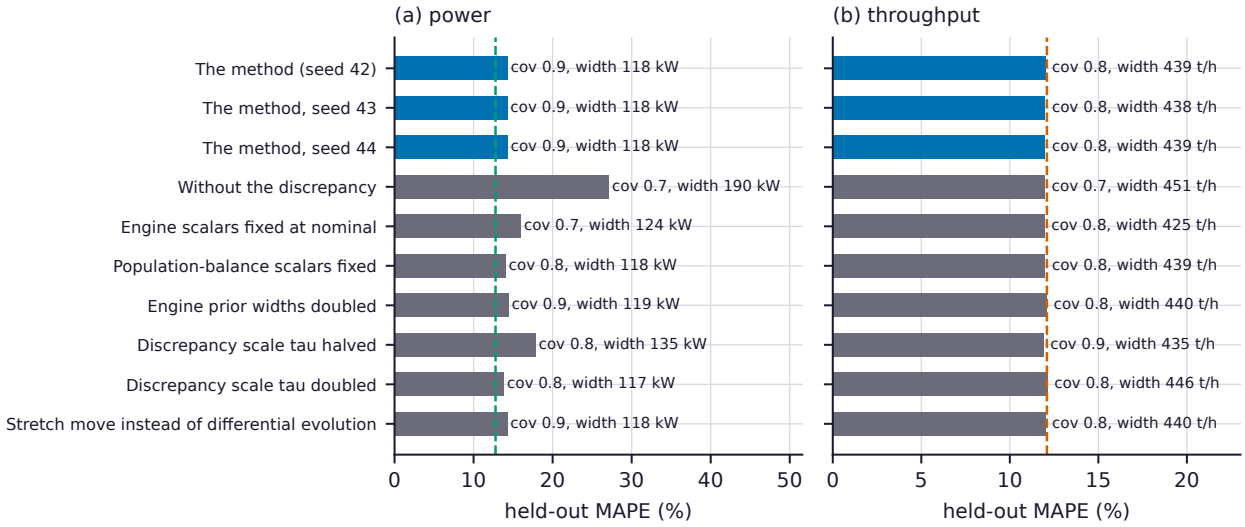


Fig. 8. Held-out MAPE of each ablation variant for (a) power and (b) throughput, with its coverage and mean band width as labels; the dashed line is the setting-only line for power and the refit capacity base for throughput, and the three seeds of the full method are in blue.

uncertainty of the scalars and of the discrepancy, and its width is set by the posterior of the noise. Against the residual band of the refit engine it gives the same coverage at 43.9 % less width on power, a gain that is plain in the data. Against the setting-only line it gives about the same width with one more survey covered out of ten, which the Wilson intervals do not resolve. Three results are negative and are part of the finding: the method does not beat a line in the setting on power, the engine-based power models (refit current-based form, Bond, Morrell through the origin, the published constants) are worse than the constant mean, and the refit current-based form on all ten surveys takes a negative slope (-2.41, with an intercept of 828.6 kW), which is the engine's power moving against the measured power. The Morrell form with a no-load term reaches 15.22 %, but its fitted work index is negative in every fold, so it is a descriptive line in the engine's size term and not an estimate of M_{ic}; through the origin its fitted M_{ic} is 3.70 to 4.21 kWh/t across folds and its error is 38.08 %. For a plant engineer the practical reading is that, on these surveys, the population balance engine contributes a calibrated throughput line and a band, and the power prediction comes from the data's own dependence on the setting.

X. LIMITATIONS

A. Internal validity

The priors and the discrepancy scales are choices: we found no published range of the fitted scalars for this ore. The ablation measures their effect (doubling the prior widths, halving and doubling τ_t) and finds that accuracy moves by at most about three and a half points on power and a quarter of a point on throughput, while the identification of the capacity base moves with τ_{tph} as the theory states; a different prior family was not tried. The convergence criterion uses the autocorrelation time of one ensemble per posterior, and the Monte Carlo error is measured by three seeds, not by independent ensembles started from dispersed points. The

linearized posterior of Section V is evaluated at one point (the posterior mean on all ten surveys) and ignores the truncation of the priors. The comparators that use the engine's P_{80} (Morrell, Bond) inherit the engine's product size, which is not measured. The seed of the bootstrap is fixed, so its intervals are reproducible.

B. External validity and construct

Ten surveys of one machine, one ore and one operation, with two measured outputs and two regressors, bound every result. Coverage is coarse: one survey moves it by 0.1, and the Wilson intervals of the coverages span about 0.5 to 0.95. The throw and the speed are fixed at the machine's nominal values, so Proposition 2 holds exactly here and would not hold with a varying speed; the feed gradation is rebuilt from the measured f_{80} with a Rosin–Rammler distribution of fixed uniformity, so neither the throw, the speed nor the feed shape is calibrated. The discrepancy is linear in two regressors with a fixed prior scale, not a Gaussian process with estimated hyperparameters, and the two outputs are independent given the scalars, although the measured power and throughput are correlated at -0.85. Whether the band stays calibrated at a setting outside the surveyed range of 38 to 55 mm was not tested. The MAPE weights a survey by the inverse of its measured value, so the low-power surveys weigh more on power. The comparison is limited to methods the product implements; discrete-element simulation is not covered, and the literature search was not exhaustive. The rule for dropping the method is lenient by construction, since it requires failure on accuracy and on calibration at once.

XI. Reproducibility

The code, data and artifacts are in the repository `github.com/fsantibanezleal/CAOS_ChancaDEM`. The posteriors of the method are sampled offline by the pipeline step of the data bake (`python data-pipeline/run.py hp500`), which writes `data/derived/hp500/crusherbayes.json`; the engine, the priors, the likelihood, the sampler and the scores are in `data-pipeline/pipeline/hp500/crusherbayes.py`, and the comparators in `data-pipeline/pipeline/hp500/loo.py`. The ablation and the seed replicates are run by `python manuscripts/crusherbayes/ablation/run_ablation.py`, which calls the same code and writes one document per variant under `manuscripts/crusherbayes/data/ablation/`; each run took 19 to 47 minutes on a workstation with 32 logical processors and 30 worker processes. The checks of the theory are computed by `python manuscripts/crusherbayes/analysis/theory_check.py` into `manuscripts/crusherbayes/data/theory.json`. Every figure, table and number of the text is regenerated from these artifacts by `python manuscripts/crusherbayes/figures/make_figs.py`, which writes the number-to-source table `manuscripts/crusherbayes/data/claims.json`. The sampler is `emcee` 3.1.6 with seed 42; the bootstrap uses seed 42. The document of the method carries the SHA-256 of its inputs,

```
483d24c5ddd35b41f2be05628fe9f09fd22562c5ab59cda69e036b9bb0ce605a
```

and the SHA-256 of the document read here is

```
f284ec33a390dbafb207c66ca7b5c6a196e746251c252958c1023682e9075778
```

XII. Conclusions

Ten HP500 surveys, two measured outputs and a linear structural discrepancy identify the capacity base, the noise and the dependence of both outputs on the setting and the feed size. They determine the work index and the power slope only as a product and leave the classification and breakage scalars of the population balance at their prior, which the theory predicts and a linearized posterior reproduces within 0.115 on every parameter. Held out, CrusherBayes reaches 12.00 % on throughput and 14.31 % on power, no better than a line in the closed-side setting (12.78 % on power, paired difference 1.54 points, interval -1.11 to 4.10), with 80 % bands that are calibrated on both outputs and 43.9 % narrower on power than the residual band of the refit engine. The ablation shows that the discrepancy carries the power prediction and that the population-balance scalars contribute nothing beyond the survey-level noise. To calibrate the population balance itself, the surveys need to measure the product size; the next steps are the digitized product size distributions of the source with their digitization error, and surveys at a varying speed, which would break the exact confounding of the capacity base.

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